

Exact Two-Dimensional $SU(2)$ Yang–Mills in Lean: Weyl Integration, Heat-Kernel Convolution, Migdal Invariance, and the Exact Simple-Loop Area Law

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13 July 2026

Abstract

We give a kernel-checked Lean 4/Mathlib formalization of the exact heat-kernel solution of two-dimensional $SU(2)$ Yang–Mills theory along the chain from Haar measure to a concrete Migdal subdivision move. The development identifies the transported Haar probability on $SU(2) \simeq S^3$ with the canonical spherical probability, proves the required all-order orbital integration law, derives translated character convolution for every pair of irreducible labels, passes from finite character sums to the infinite heat-kernel semigroup by dominated convergence, and integrates an actual edge shared by two faces. Combined with the all-label coefficient extraction already proved in the same development, this yields the exact normalized simple-loop identity

$$\mathbb{E}[W_n(C)] = \exp\left[-\frac{n(n+2)}{4} \text{Area}(C)\right].$$

All displayed endpoints compile without `sorry`, `admit`, or project-local axioms. The mathematical identities are classical; the claimed contribution is their concrete end-to-end machine-checked composition, not a new analytic solution of two-dimensional Yang–Mills theory.

1 Statement and claim boundary

Let dg be normalized Haar probability on $SU(2)$, let χ_n be the character of the irreducible representation of dimension $d_n = n + 1$, and put

$$c_n = \frac{n(n+2)}{4}, \quad K_t(g) = \sum_{n \geq 0} d_n e^{-tc_n} \chi_n(g), \quad t > 0.$$

With convolution

$$(f * h)(g) = \int_{SU(2)} f(x) h(x^{-1}g) dx,$$

the formalized chain is

$$\begin{aligned} \text{Haar on } S^3 &\implies \text{all-order orbital law} \implies \chi_n * \chi_m \\ &\implies K_s * K_t = K_{s+t} \implies \text{Migdal} \implies \text{simple-loop coefficient}. \end{aligned}$$

This result is distinct from the programme’s earlier strong-coupling area-law theorem. The earlier theorem is a volume-uniform *inequality* for finite lattices and $SU(N_c)$ in an explicit convergence window, obtained from a Kotecký–Preiss cluster expansion. The present theorem is an *exact identity* in the two-dimensional $SU(2)$ heat-kernel model, obtained by Haar

integration and subdivision. The former is cited context; it is not used to discharge any exact two-dimensional statement here.

No claim is made here about four-dimensional Yang–Mills, a continuum construction in four dimensions, or a new proof of the classical exact solution. A priority claim about formalization should remain conditional on a systematic literature and repository search.

2 From spherical Haar measure to the orbital law

Write an element of $SU(2)$ in its standard form

$$g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}, \quad |a|^2 + |b|^2 = 1.$$

Earlier modules construct a homeomorphism $SU(2) \simeq S^3$, realize left multiplication as an ambient real-linear isometry, prove preservation of the canonical `Measure.toSphere` measure, and use uniqueness of normalized Haar measure to obtain literal equality of the two probability measures.

The new all-order step studies $u = |a|^2$. In dimension four, the squared mass in one complex rail is uniform on $[0, 1]$. Rather than assume this distribution, the formal proof establishes its moment recurrence, proves

$$\int_{SU(2)} u^k dg = \frac{1}{k+1} \quad (k \geq 0),$$

and then identifies measures on the compact interval using polynomial density. The resulting literal endpoint is:

```
theorem su2FirstRailMassMeasure_eq_uniform :
  su2FirstRailMassMeasure = su2UnitIntervalMeasure
```

For $x, y \in [-1, 1]$, choose representatives with half-traces x, y and average their product over conjugation. The preceding uniform law reduces the orbital integral to a one-variable Chebyshev integral. For the Chebyshev polynomial U_m of the second kind, the formalized formula is

$$\int_{SU(2)} U_m\left(\frac{1}{2} \operatorname{tr}(kak^{-1}b)\right) dk = \frac{U_m(\frac{1}{2} \operatorname{tr} a) U_m(\frac{1}{2} \operatorname{tr} b)}{m+1}.$$

The proof includes the endpoint cases $x = \pm 1$; no division by a vanishing sine is hidden in the statement.

3 All-order translated character convolution

Characters are implemented concretely as

$$\chi_n(g) = U_n\left(\frac{1}{2} \operatorname{tr} g\right).$$

Conjugacy invariance, left and right Haar invariance, Fubini, and the orbital law give the translated orthogonality relation

$$\int_{SU(2)} \chi_n(x) \chi_m(x^{-1}g) dx = \delta_{nm} \frac{\chi_n(g)}{n+1}. \quad (1)$$

This is stronger than orthogonality at the identity and is exactly the consumer needed by the heat-kernel product.

Theorem 3.1 (Machine-checked character convolution). *For every $n, m \in \mathbb{N}$ and $g \in SU(2)$,*

$$(\chi_n * \chi_m)(g) = \begin{cases} \chi_n(g)/(n+1), & n = m, \\ 0, & n \neq m. \end{cases}$$

The public Lean endpoints are listed below. They are theorems about the actual normalized Haar integral, not fields projected from an abstract character package.

```
integral_su2CharacterReal_mul_translate
su2CharacterChebyshev_convolution
```

4 Infinite heat-kernel semigroup

For a cutoff N , expand both finite kernels, interchange the finite sums and the Haar integral, and apply (1). The Kronecker selector leaves a single sum and the Casimir weights multiply:

$$e^{-sc_n} e^{-tc_n} = e^{-(s+t)c_n}.$$

This proves the finite convolution identity. The infinite theorem then uses the already established uniform Casimir majorant for positive times. Partial kernels converge pointwise and uniformly; the product is bounded by the product of two summable majorants, so dominated convergence passes the Haar integral to the limit.

Theorem 4.1 (Concrete heat semigroup). *For $s, t > 0$ and every $g \in SU(2)$,*

$$\int_{SU(2)} K_s(x) K_t(x^{-1}g) dx = K_{s+t}(g).$$

*Equivalently, as functions, $K_s * K_t = K_{s+t}$.*

The corresponding Lean declarations, including the function-level consumer, are:

```
su2HeatKernelPartial_convolution
su2HeatKernel_convolution
su2HeatKernel_convolution_consumer
```

5 A concrete Migdal move

Consider two adjacent faces of areas s and t . Let a, b encode the external boundary holonomies and integrate their shared oriented edge x . The actual two-face density in the development is

$$D_{s,t}^{a,b}(x) = K_s(ax) K_t(x^{-1}b).$$

Left invariance changes variables $y = ax$ and the heat semigroup gives

$$\int_{SU(2)} D_{s,t}^{a,b}(x) dx = K_{s+t}(ab). \quad (2)$$

Theorem 5.1 (Two-face Migdal subdivision). *Equation (2) holds for all $a, b \in SU(2)$ and all $s, t > 0$. It is proved by a genuine Haar integration over the shared edge and invokes the concrete infinite heat-kernel semigroup.*

This closes the first nontrivial subdivision move: two holonomy variables, two faces, and one integrated shared edge. The endpoints below contain no package hypothesis.

```
su2Migdal_twoFace_merge
su2Migdal_subdivision_invariant
```

6 Exact simple-loop coefficient

The same repository proves, for every label n and positive area t , that the normalized Wilson character $W_n = \chi_n/(n+1)$ extracts the corresponding heat coefficient:

$$\int_{SU(2)} W_n(g) K_t(g) dg = e^{-tc_n} = \exp\left[-t \frac{n(n+2)}{4}\right].$$

The proof first establishes the identity for partial character sums and then passes to the full kernel by dominated convergence. Its final consumer is:

```
theorem su2_exact_simpleLoop_areaLaw
  {t : Real} (ht : 0 < t) (n : Nat) :
  (integral (fun g : SU2 =>
    su2NormalizedWilsonCharacter n g *
    heatKernelCharacterSeries su2CharacterTable t g)
    su2HaarProb) =
  ((Real.exp (-t * ((n : Real) *
    ((n : Real) + 2) / 4)) : Real) : Complex)
```

Together with the Migdal move, this is the exact two-dimensional route: edge subdivision does not change the density, and the remaining simple face has the Casimir coefficient prescribed by its area.

7 Audit and reproducibility

The checked source is built with Lean 4 and Mathlib. The public root module imports the concrete orbit, character-convolution, heat-semigroup, and exact area-law modules. A repository-wide placeholder audit finds no occurrence of `sorry`, `admit`, or a project-local `axiom` declaration in the Lean development. Printing axioms for the twelve new public endpoints returns only the standard kernel dependencies

`propxt,      Classical.choice,      Quot.sound.`

The toolchain and Mathlib revision are pinned in the repository. GitHub Actions builds the public root target from a clean checkout and separately rejects placeholders.

Principal new declarations.

```
su2FirstRailMoment_recurrence
su2FirstRailMoment_eq
su2FirstRailMassMeasure_eq_uniform
intervalIntegral_chebyshevU_product_formula
integral_su2FirstEntryMass_eq_unitInterval
integral_su2Haar_orbit_character_general
integral_su2CharacterReal_mul_translate
su2CharacterChebyshev_convolution
su2HeatKernelPartial_convolution
su2HeatKernel_convolution
su2Migdal_twoFace_merge
su2Migdal_subdivision_invariant
```

8 Scope and next frontier

The present artifact closes the compact-group analytic chain needed for an exact simple-loop calculation. It does not yet formalize arbitrary embedded planar graphs, gauge fixing on a

general cellular decomposition, nonsimple or intersecting loops, continuum measure construction, or higher-genus partition functions. Those are independent extensions and should not be inferred from the two-face theorem.

The next mathematically meaningful consumer is a finite planar-cellulation reduction theorem: repeated application of the concrete Migdal move, with incidence and orientation data made explicit, until a simple loop bounds one effective face. That step would promote the local subdivision identity into a global discretization-independence statement.

References

- [1] A. A. Migdal, *Recursion equations in gauge field theories*, Sov. Phys. JETP **42** (1975), 413–418.
- [2] E. Witten, *On quantum gauge theories in two dimensions*, Commun. Math. Phys. **141** (1991), 153–209.
- [3] B. K. Driver, *Yang–Mills theory and the heat equation on compact Lie groups*, J. Funct. Anal. **110** (1992), 1–34.
- [4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, Lecture Notes in Computer Science 12699 (2021), 625–635.
- [5] The Mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.